Exp. 1
Date: 20/1/24

Study of various Network commands Used in linux and windows

Basic Networking commands

1. arp -a : ARP is short form of address resolution protocol, It will show the IP address of your computer along with the IP address and MAC address of your router.

Output:
-gateway (10.0.2.2) at 52:54:00:12:35:02 [ether] on eth0

2. hostname : This is the simplest of all TCP/IP commands. It is simply displays the name of your computer

Output:
localhost.localdomain

3. Ipconfig /all : This command displays detailed configuration information about your TCP/IP connection including Router, Gateway, DNS, DHCP and type of Ethernet adapter in your system

Output:
Windows IP configuration

Ethernet adapter Ethernet 2:

Connection-Specific DNS Suffix :
Link-Local IPV6 Address . . . : fe80::6040:73f8:684d:a4be
IPV4 Address . . . . . : 192.168.56.1
Subnet Mask . . . . . : 255.255.255.0

4. Nbtstat -a : This command helps solve problems with NetBIOS name resolution.

Output:
Displays protocol statistics and current TCP/IP connections using NBT

NBTSTAT [[-a RemoteName] [-A IP Address] [-c] [n] [-r] [-R] [-RR] [-s] [-S] [Interval]]

5. netstat: netstat displays a variety of statistics about a computer active TCP/IP connections. It is a command line tool for monitoring network connections both incoming and outgoing as well as viewing routing tables, interface statistics etc.

Output:

Active connections

Proto	Local Address	Foreign Address	State
TCP	10.55.47.189:50357	4:213.25.240:hHps	Established
TCP	10:55:47.189:50358	4:213.25.240:hHps	Established

6. nslookup: It is a tool used to perform DNS lookups in linux. It is used to display DNS details, such as the IP address of a particular computer the MX record for a domain.

Output:

Default server: Unknown
Address: 10.55.47.248

7. PathPing: Pathping is a unique to windows and it is basically a combination of ping and Tracert command

Output

Usage: Pathping [-g host-list] [-h maximum-hops] [-i address] [-n] [-p period] [-q num-queries] [-w timeout] [-4] [-6] target-name

8. Ping: (Packet INternet Groper) command is the best way to test connectivity between two nodes.

Output:

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS] [-r count] [-s count] [-j host-list] [-k host-list]

9. Route: Route command is used to show/manipulate the IP routing table.

Output:

Kernel IP routing table

Destination	Gateway	Genmask	Flags	Metric	Ref	Use	Iface
default	-gateway	0.0.0.0	UG	100	0	0	enp0s3
10.0.2.0	0.0.0.0	255.255.255.255	U	100	0	0	enp0s

Some important linux networking commands

1. ip: The ip command can show address information, manipulate routing, plus displays network various devices, interface and tunnels.

i) ip address show

Output:

1: lo: <LOOPBACK, UP, LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlean 1000

ii) To assign an IP to an interface for example enp0s3

2. ifconfig

The ifconfig command was/is a staple in many sysadmin's tool belt for configuring and troubleshooting between network

Output:

enp0s3: flags=4163<UP, LOOPBACK, RUNNING, MULTICAST>
mtu 1500 inet 10.0.2.15 netmask 255.255.255.0 broadcast
10.0.2.255 inet6 fe80::a00:27ff:fead:4946 prefixlen 64
scopeid 0x20<link>

3. mtr

MTR is a program with a command line interface that serves as a network diagnostic and troubleshooting tools. This command combines the functionality of the Ping and troubleshooting tool.

Output

My traceroute [V0.94]

Host	Packets		Pings				
1. Localhost	loss%	Snt	Last	Avg	Best	Wrst	StDev
	0.0%	122	0.1	0.2	0.1	1.9	0.2

ii) mtr. google.com

Output host

1. gateway
2. (waiting for reply)

iii) mtr. -g google.com

Output

Usage:

mtr [options] hostname

-F, --filename FILE read host name(s) from a file.

4. tcpdump The tcpdump command is designed for capturing and displaying packets.

Output:

tcpdump: enp0s3: you don't have permission to capture on that device.

5 Ping : Ping is a tool that verifies IP-level connectivity to another TCP/IP computer by sending Internet Control Message Protocol (ICMP) Echo Request messages.

vi; Ping google.com

Output:

PING google.com (142.250.182.78) 56(84) bytes of data

64 bytes from lcmaaa-ax-in-f14.1e100.net (142.250.182.78):

icmp-seq=1. ttl=112 time=938 ms.

Result:

Thus the various network commands used in Linux and windows was executed successfully.